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Title:

Demonstrate the use of Logistic Regression on data sets used in Linear Regression and compare with results of linear regression

AIM:

To demonstrate the use of Logistic Regression on data sets used in Linear Regression and compare with results of linear regression

Steps:

1.Import required packages.

2.Load and prepare the dataset.

3.Split data into training and testing sets.

4.Apply feature scaling.

5.Create and train the Logistic Regression model.

6.Predict and evaluate the model performance.

Source Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
data = pd.read_csv("test.csv")
data['battery_class'] = pd.qcut(data['battery_power'], q=3,labels=False)
```

```
data = data.dropna()
X = data.drop(columns=['battery_power','battery_class']).to_numpy()
y = data['battery_class'].to_numpy()
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)
cr = classification_report(y_test, y_pred)
print(f"Accuracy: {accuracy*100:.2f}%")
print("\nConfusion Matrix:\n", cm)
print("\nClassification Report:\n", cr)
plt.figure(figsize=(7,5))
plt.scatter(X[:,0], X[:,1], c=y, cmap='viridis', edgecolor='k')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("Scatter Plot")
plt.colorbar(label="Target")
plt.show()
plt.figure(figsize=(10,6))
plt.boxplot(X)
plt.xlabel("Feature Index")
plt.ylabel("Value")
plt.title("Box Plot of All Features")
```

plt.show()

INPUTS:test.csv

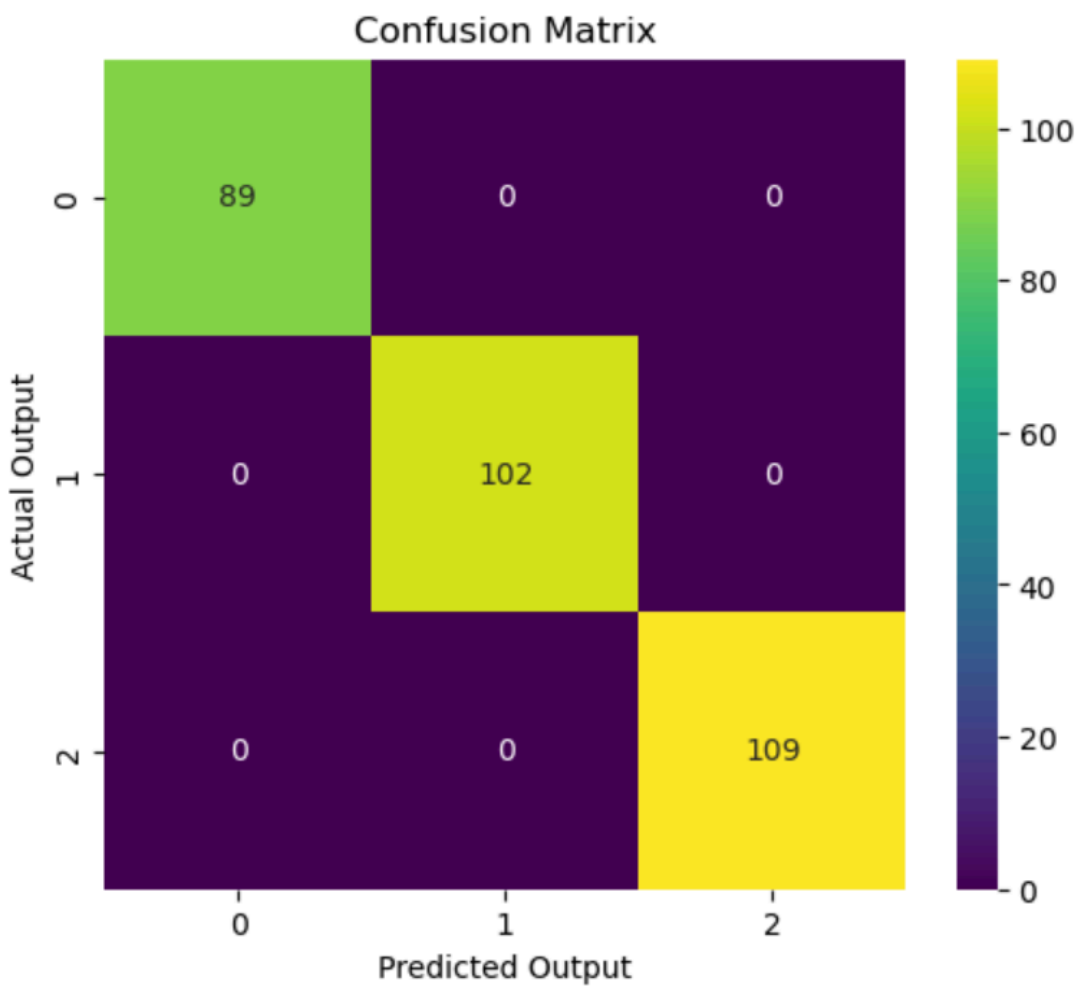
OUTPUTS:

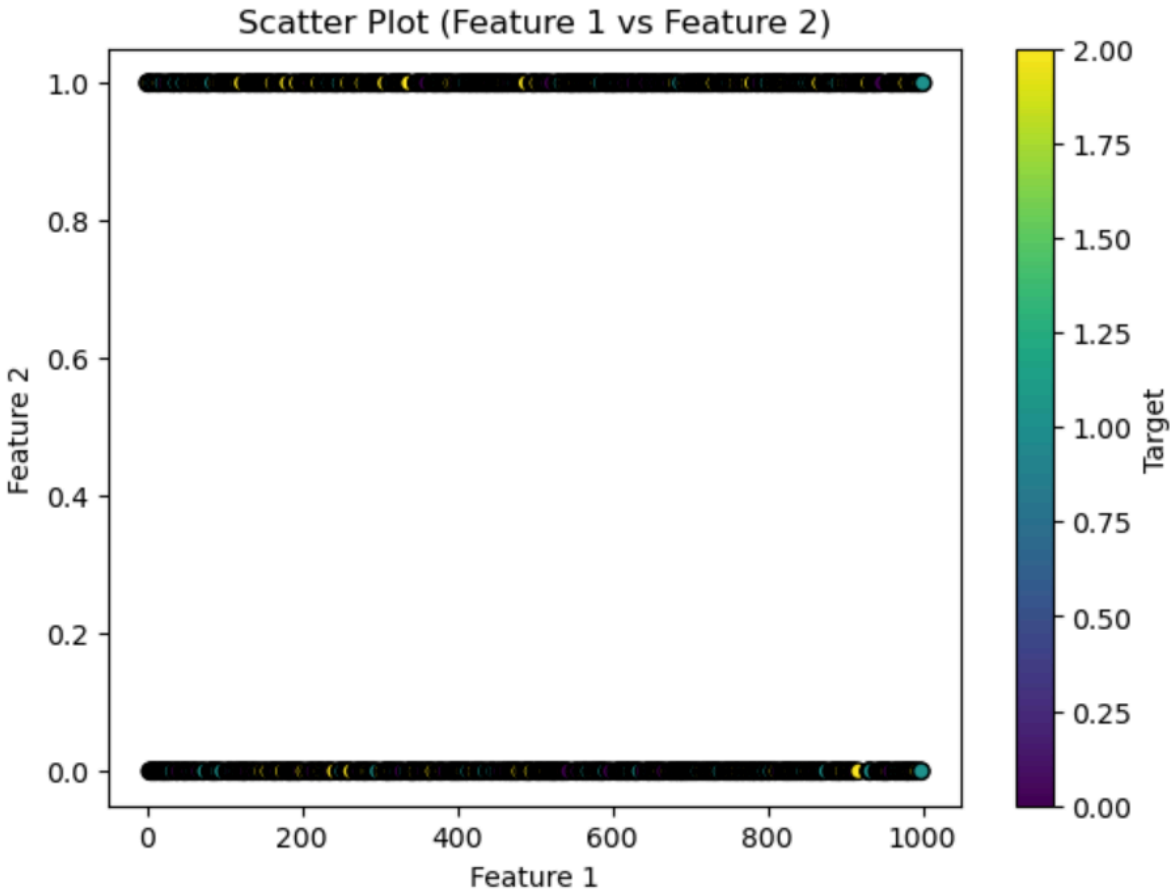
1)Confusion matrix

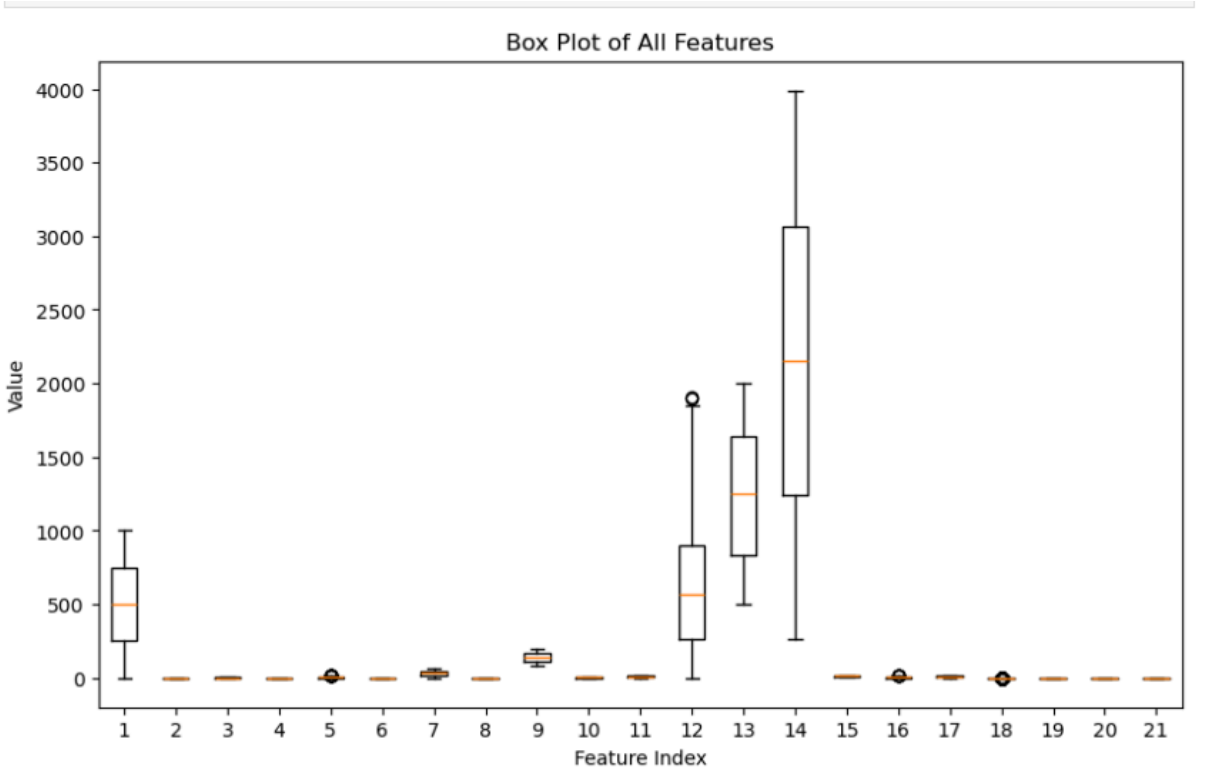
2)Scatter plot

3)Box plot

ScreenShots:




Scatter Plot (Feature 1 vs Feature 2)
Feature 2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
Feature 1
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Target



Key Observations:

1) The Logistic Regression model classified the data into different categories based on the input features.
2)It uses the sigmoid function to calculate probabilities and then assigns the class with the highest probability.
3)The model performance was checked using Accuracy, Confusion Matrix, and Classification Report.
4)Some errors happened because some data values were similar or overlapping.
5)The model worked better when the data was clearly separated and properly scaled using StandardScaler.

Result:

The Logistic Regression model worked successfully and predicted the categories based on the given input data.